OOP Expense single data capturing

DB Schema

```sql
CREATE TABLE `opt_pocket_expense_type` (
  `id` INT(11) UNSIGNED NOT NULL AUTO_INCREMENT,
  `option` VARCHAR(128) NOT NULL DEFAULT '',
  `amount_sign` ENUM('positive','negative') DEFAULT 'negative',
  PRIMARY KEY (`id`)
);

INSERT INTO `opt_pocket_expense_type` (`option`, `amount_sign`) VALUES
('ATM Withdrawal', 'negative'),
('Point of Sale', 'negative'),
('Fee & Charges', 'negative'),
('Refund from Merchant', 'positive');

Source :
CREATE TABLE `pocket_expense_source_client_config` (
  `id` INT(10) UNSIGNED NOT NULL AUTO_INCREMENT,
  `uuid` VARCHAR(36) DEFAULT NULL,
  `client_id` INT(10) UNSIGNED DEFAULT NULL,
  `name` VARCHAR(100) NOT NULL,
  `is_default` TINYINT(1) UNSIGNED NOT NULL DEFAULT 0,
  `deleted` TINYINT(1) UNSIGNED NOT NULL DEFAULT 0,
  `delete_time` DATETIME DEFAULT NULL,
  `create_time` DATETIME DEFAULT NULL,
  `update_time` DATETIME DEFAULT NULL,
  PRIMARY KEY (`id`),
  UNIQUE KEY `oop_expense_source_client_name_unique` (`client_id`, `name`),
  CONSTRAINT `oop_expense_source_client_id_foreign` FOREIGN KEY (`client_id`) REFERENCES `client` (`id`)
);

CREATE TABLE `pocket_expense` (
  `id` INT(10) UNSIGNED NOT NULL AUTO_INCREMENT,
  `uuid` VARCHAR(36) DEFAULT NULL,
  `user_id` INT(11) UNSIGNED NOT NULL,
  `client_id` INT(11) UNSIGNED DEFAULT NULL,
  `date` DATE NOT NULL,
  `merchant_name` VARCHAR(180) NOT NULL,
  `merchant_description` VARCHAR(255) DEFAULT NULL,
  `expense_type` INT(11) UNSIGNED DEFAULT NULL,
  `currency` CHAR(3) NOT NULL,
  `amount` DECIMAL(14,2) NOT NULL DEFAULT 0.00,
  `merchant_address` VARCHAR(500) DEFAULT NULL,
  `vat_amount` DECIMAL(12,2) DEFAULT NULL,
  `notes` TEXT DEFAULT NULL,
  `status` ENUM('draft','submitted','approved','rejected') DEFAULT 'draft',
  `created_by_user_id` INT(11) UNSIGNED NOT NULL,
  `updated_by_user_id` INT(11) UNSIGNED DEFAULT NULL,
  `approved_by_user_id` INT(11) UNSIGNED DEFAULT NULL,
  `create_time` DATETIME DEFAULT CURRENT_TIMESTAMP,
  `update_time` DATETIME DEFAULT NULL ON UPDATE CURRENT_TIMESTAMP,
  `deleted` TINYINT(1) UNSIGNED NOT NULL DEFAULT 0,
  `delete_time` DATETIME DEFAULT NULL,

  PRIMARY KEY (`id`),
  KEY `pocket_expense_user_id_foreign` (`user_id`),
```

```
  KEY `pocket_expense_client_id_foreign` (`client_id`),
  KEY `pocket_expense_expense_type_foreign` (`expense_type`),
  CONSTRAINT `pocket_expense_user_id_foreign` FOREIGN KEY (`user_id`) REFERENCES `user` (`id`),
  CONSTRAINT `pocket_expense_client_id_foreign` FOREIGN KEY (`client_id`) REFERENCES `client` (`id`),
  CONSTRAINT `pocket_expense_expense_type_foreign` FOREIGN KEY (`expense_type`) REFERENCES `opt_pocket_expense_type` (`id`)
) ;




CREATE TABLE `pocket_expense_metadata` (
  `id` INT(10) UNSIGNED NOT NULL AUTO_INCREMENT,
  `pocket_expense_id` INT(10) UNSIGNED NOT NULL,
  `metadata_type` ENUM('category','tracking_code_type_1','tracking_code_type_2','project','additional_field','file','expense_source') NOT NULL,

  -- Dedicated Foreign Key Columns (for fast joins)
  `transaction_category_id` INT(10) UNSIGNED DEFAULT NULL,
  `tracking_code_id` INT(10) UNSIGNED DEFAULT NULL,
  `project_id` INT(10) UNSIGNED DEFAULT NULL,
  `file_store_id` INT(10) UNSIGNED DEFAULT NULL,
  `expense_source_id` INT(10) UNSIGNED DEFAULT NULL,
  `additional_field_id` INT(10) UNSIGNED DEFAULT NULL,

  -- User reference
  `user_id` INT(11) UNSIGNED NOT NULL,

  -- JSON for all details
  `details_json` JSON DEFAULT NULL,

  `create_time` DATETIME DEFAULT CURRENT_TIMESTAMP,
  `update_time` DATETIME DEFAULT NULL ON UPDATE CURRENT_TIMESTAMP,
  `deleted` TINYINT(1) UNSIGNED NOT NULL DEFAULT 0,
  `delete_time` DATETIME DEFAULT NULL,

  PRIMARY KEY (`id`),

  -- Unique constraints (one record per type per expense)
  UNIQUE KEY `metadata_category_unique` (`pocket_expense_id`, <category>, `deleted`),
  UNIQUE KEY `metadata_tracking_type_1_unique` (`pocket_expense_id`, `metadata_type`, `deleted`),
  UNIQUE KEY `metadata_tracking_type_2_unique` (`pocket_expense_id`, `metadata_type`, `deleted`),
  UNIQUE KEY `metadata_project_unique` (`pocket_expense_id`, `metadata_type`, `deleted`),
  UNIQUE KEY `metadata_source_unique` (`pocket_expense_id`, `metadata_type`, `deleted`),
  ==Q=> why multiple keys with same constrains?

  -- Foreign Keys
  CONSTRAINT `metadata_expense_id_foreign` FOREIGN KEY (`pocket_expense_id`) REFERENCES `pocket_expense` (`id`),
  CONSTRAINT `metadata_category_id_foreign` FOREIGN KEY (`transaction_category_id`) REFERENCES `transaction_category` (`id`),
  CONSTRAINT `metadata_tracking_code_id_foreign` FOREIGN KEY (`tracking_code_id`) REFERENCES `tracking_codes` (`id`),
  CONSTRAINT `metadata_project_id_foreign` FOREIGN KEY (`project_id`) REFERENCES `configurable_projects` (`id`),
  CONSTRAINT `metadata_file_store_id_foreign` FOREIGN KEY (`file_store_id`) REFERENCES `file_store` (`id`),
  CONSTRAINT `metadata_source_id_foreign` FOREIGN KEY (`expense_source_id`) REFERENCES `pocket_expense_source_client_config` (`id`),
  CONSTRAINT `metadata_additional_field_id_foreign` FOREIGN KEY (`additional_field_id`) REFERENCES `expense_additional_field` (`id`),
  CONSTRAINT `metadata_user_id_foreign` FOREIGN KEY (`user_id`) REFERENCES `user` (`id`),
```

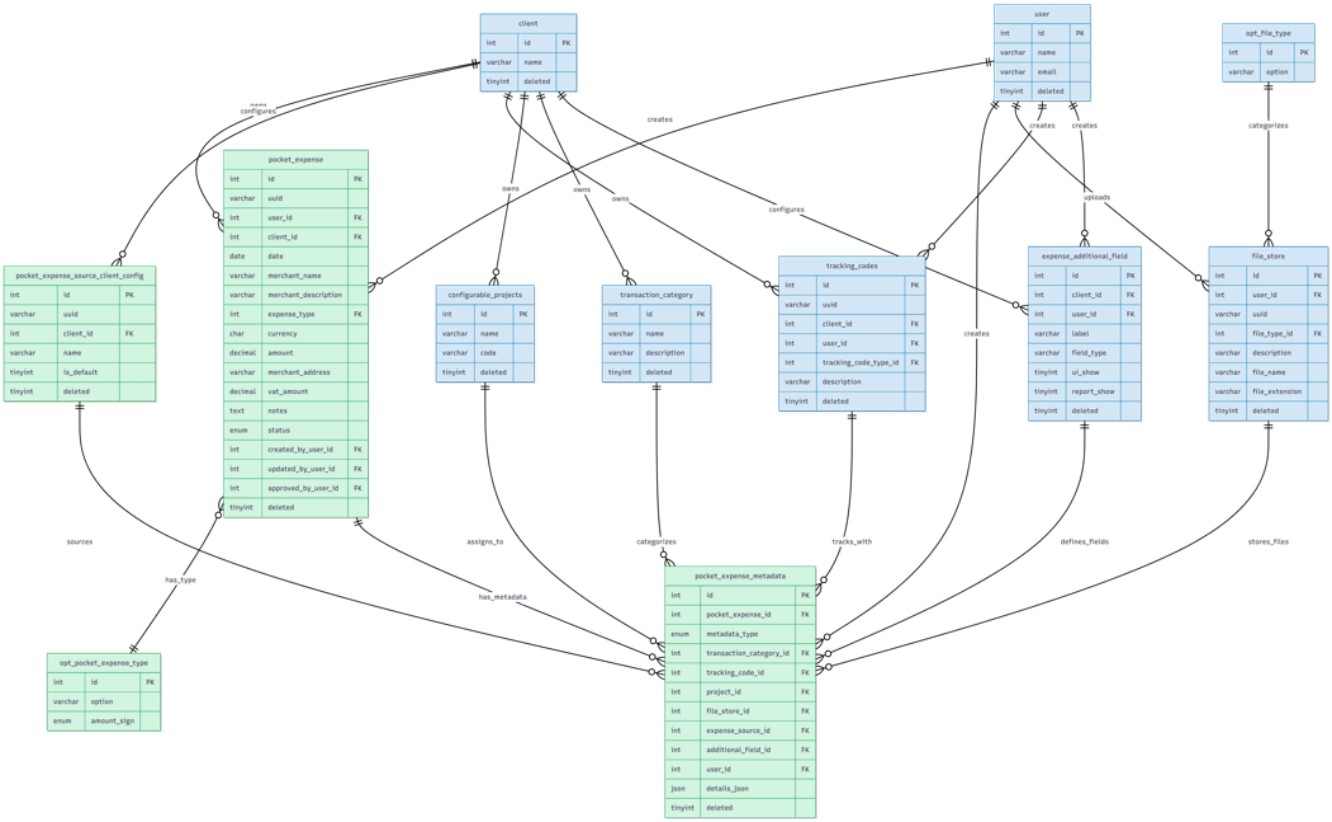
```

  -- Indexes
  KEY `metadata_expense_id_index` (`pocket_expense_id`),
  KEY `metadata_type_index` (`metadata_type`),
  KEY `metadata_category_index` (`transaction_category_id`),
  KEY `metadata_tracking_code_index` (`tracking_code_id`),
  KEY `metadata_project_index` (`project_id`),
  KEY `metadata_file_index` (`file_store_id`),
  KEY `metadata_source_index` (`expense_source_id`)
)

JSON Example
1. Category Metadata
json
{
  "other": "Custom Software Development",
  "is_other": 1,
  "field_1_value": "Engineering Department",
  "field_2_value": "Project Alpha Budget"
}
2. Additional Field Metadata
json
{
  "value": "Client meeting and product demo"
}
3. Expense Source Metadata (for custom sources)
json
{
  "expense_source_other_value": "blah blah PayPal"
}
4. Tracking Code Metadata
json
{
}
5. Project Metadata
json
{
}
5.File Metadata
json
{
}
```

🟢 Green: New tables

🔵 Blue: existing tables



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OOP Expense source

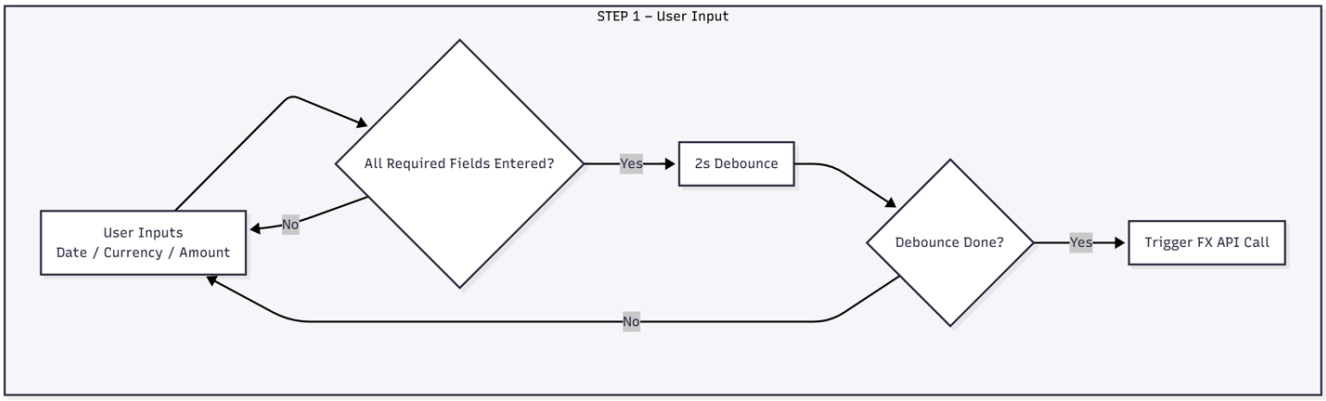
- **Default and Global Source Setup**
  - When a client enables the OOP feature:
    - Automatically create 3 default sources for that client: *Cash*, *Corporate Card*, *Personal Card* when oop feature is enabled on table oop_expense_source_client_config
    - Ensure a single global "Other" record exists in the master table (`client_id = NULL`). (add into the migration)
  - "Other" is not deletable or editable by clients.
- **Dropdown Display**
  - Dropdown in the OOP expense form should list:
    - All **active client-specific sources** (`client_id = current_client` , `deleted = 0`)
    - The **global "Other"** source (`client_id IS NULL`).

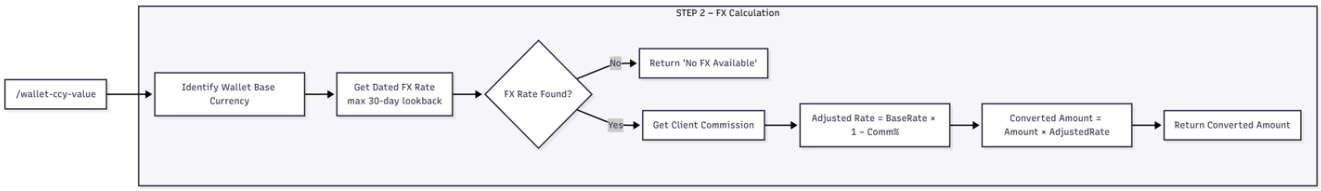
- **Expense Submission**
  - When user selects a source (non-“Other”):
    - Save `expense_source_id` only.
  - When user selects “Other”:
    - Save `expense_source_id` = ID of global “Other” record.
    - Save user input text into `custom_source_text` field.
- **Client Config Behaviour under expense tag**
  - Maximum of 20 active (non-deleted) sources allowed per client.
  - Client may soft-delete any client-specific source (set `deleted = 1`).
  - Deleted sources:
    - Remain visible on historical expense details.
    - Are excluded from future dropdowns.
- **Validation Rules**
  - Prevent creation of more than 20 active sources per client.
  - Enforce unique source names per client (`client_id`, `name`).

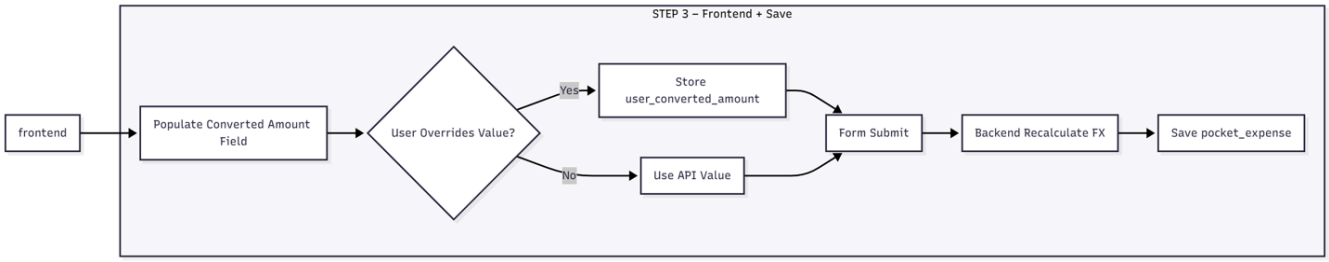
Currency Conversion

possible query:

```
SELECT
    pc.client_id,
    acc.base_currency_id as account_base_currency_id,
    acc_curr.currency as account_base_currency_code,
    atier.base_currency_id as tier_base_currency_id,
    tier_curr.currency as tier_base_currency_code,
    pc.card_currency
FROM prepaid_card pc
JOIN prepaid_card_tier_map pctm ON pctm.prepaid_card_id = pc.id
JOIN account_tier atier ON atier.id = pctm.account_tier_id
JOIN account acc ON acc.id = pc.account_id
JOIN currency acc_curr ON acc_curr.id = acc.base_currency_id
JOIN currency tier_curr ON tier_curr.id = atier.base_currency_id
WHERE pc.client_id = 213321
    AND atier.master_virtual_card = 1
    AND pc.deleted = 0;
```








```
flowchart LR

%% ---------- STEP 1 ----------
subgraph STEP_1[STEP 1 – User Input]
direction LR
A[User Inputs<br/>Date / Currency / Amount]
B{All Required Fields Entered?}
C[2s Debounce]
D{Debounce Done?}
E[Trigger FX API Call]
end

A --> B
B -->|Yes| C
B -->|No| A
C --> D
D -->|Yes| E
D -->|No| A
```

```
flowchart LR

%% ---------- STEP 2 ----------
subgraph STEP_2[STEP 2 – FX Calculation]
direction TB
F[Identify Wallet Base Currency]
G[Get Dated FX Rate<br/>max 30-day lookback]
H{FX Rate Found?}
I[Return 'No FX Available']
J[Get Client Commission]
K[Adjusted Rate = BaseRate × 1 - Comm%]
L[Converted Amount = Amount × AdjustedRate]
M[Return Converted Amount]
end
```

```

/wallet-ccy-value --> F
F --> G
G --> H
H -->|No| I
H -->|Yes| J
J --> K
K --> L
L --> M
```

```
flowchart LR

%% ---------- STEP 3 ----------
subgraph STEP_3[STEP 3 - Frontend + Save]
direction TB
N[Populate Converted Amount Field]
O{User Overrides Value?}
P[Store user_converted_amount]
Q[Use API Value]
R[Form Submit]
S[Backend Recalculate FX]
T[Save pocket_expense]
end

frontend --> N
N --> O
O -->|Yes| P
O -->|No| Q
P --> R
Q --> R
R --> S
S --> T
```